

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_no_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_no_replay.

Run Metadata

- run_name: run_20260513_192447_h
- started_at_local: 2026-05-13 19:24:47
- finished_at_local: 2026-05-13 19:27:23
- duration_hhmm: 00:03
- duration_seconds: 156.231
- seed_offset: 7000
- replicate_label: h
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.218845	0.005376	0.12397	0.742409	0.78	0.035508
cvrp_random_replay	random	score_only	False	0.241545	0.005376	0.136581	0.727318	0.58	0.002027
cvrp_failure_replay	failure	score_only	False	0.24828	0.005376	0.140323	0.762695	0.58	0.05248
cvrp_failure_replay_compression	failure	novelty_gate	True	0.22084	0.026996	0.134687	0.780645	0.78	0.009335

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.218845, synthetic 0.005376, combined 0.12397.

- Accepted-epoch count 3, mean accepted code novelty 0.742409, and final complexity 0.78.
- Adaptation efficiency 0.035508 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_cvrplib mean gap 0.218845 across 5 instances; family means: A=0.091623, B=0.243412, E=0.264875, P=0.250903.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 937, "cost": 1280, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.366062}, {"best_known_cost": 458, "cost": 618, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.349345}, {"best_known_cost": 672, "cost": 845, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.25744}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.241545, synthetic 0.005376, combined 0.136581.
- Accepted-epoch count 2, mean accepted code novelty 0.727318, and final complexity 0.58.
- Adaptation efficiency 0.002027 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.241545 across 5 instances; family means: A=0.103839, B=0.251508, E=0.299424, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 591, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.290393}, {"best_known_cost": 937, "cost": 1065, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.136606}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.24828, synthetic 0.005376, combined 0.140323.
- Accepted-epoch count 2, mean accepted code novelty 0.762695, and final complexity 0.58.
- Adaptation efficiency 0.05248 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.24828 across 5 instances; family means: A=0.13438, B=0.252174, E=0.299424, P=0.303249.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 591, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.290393}, {"best_known_cost": 937, "cost": 1072, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.144077}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.22084, synthetic 0.026996, combined 0.134687.

- Accepted-epoch count 2, mean accepted code novelty 0.780645, and final complexity 0.78.
- Adaptation efficiency 0.009335 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.22084 across 5 instances; family means: A=0.147469, B=0.272493, E=0.245681, P=0.166065.
- Panel synthetic_holdout mean gap 0.026996 across 4 instances; family means: alternating_belt=0.064516, clustered_demand=0.043468, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 458, "cost": 588, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.283843}, {"best_known_cost": 672, "cost": 805, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.197917}]

Judge Appendix

Summary of CVRP Benchmark Results

Condition	CVRPL IB Gap	Synthetic Gap	Transfer Gap	Code Novelty (mean)	Replay Mode	Notes on Complexity & Archive
cvrp_no_replay	0.2188	0.0054*	0.1240**	0.7424	none	Complexity 0.78, Archive size 14
cvrp_random_replay	0.2415	0.0054	0.1366	0.7273	random	Complexity 0.58, Archive size 14
cvrp_failure_replay	0.2483	0.0054	0.1403	0.7627	failure	Complexity 0.58, Archive size 14
cvrp_failure_replay_compression	0.2208	0.0270	0.1347	0.7806	failure + compression	Complexity 0.78, Archive size 14

Key Interpretations

- **Best performance in transfer and held-out gaps occurs under the no replay condition (cvrp_no_replay):**
- Exhibits lowest mean gaps on held-out CVRPLIB (0.2188) and synthetic data (0.0054).
- Also has the best mean transfer gap (0.1240).
- **Replay usage (random or failure) generally increases both CVRPLIB and transfer gaps despite slight variations:**
- Random replay and failure replay conditions show worse performance on main transfer metrics (+0.02–0.03 gap increase in CVRPLIB and ~0.01–0.02 in transfer gap).
- Compression-aware failure replay improves synthetic gap slightly versus failure replay without compression (0.027 vs. 0.0054 is actually worse in synthetic gap), but still lags behind no replay.
- **Code novelty is slightly decreased in random replay (0.727) but higher for failure replay variants (0.763–0.781):**
- Despite higher code novelty under failure replay conditions, transfer metrics degrade compared to no replay.
- This suggests **code novelty improvements do not translate into better transfer** in failure or random replay cases.
- **Complexity differences:**

- No replay and failure replay with compression maintain higher complexity (~0.78) while other replay modes are less complex (~0.58).
- Complexity does not align consistently with gap improvements.
- **Family-wise worst cases:**
- Larger gaps on families B, E, and P across all conditions.
- No replay condition has slightly lower worst-case optimality gaps.

Conservative Conclusions

1. **No replay yields the best transfer and heldout optimality gaps despite slightly lower code novelty than failure-replay variants.** This prioritizes the transfer metrics over mere code novelty or complexity.
2. **Random and failure replay approaches lead to degraded transfer performance, even if code novelty is marginally higher, indicating replay strategies here do not improve cross-distribution generalization.**
3. **Compression-aware failure replay improves complexity and code novelty but does not outperform no replay in transfer gaps nor synthetic gaps, suggesting compression pressure alone is insufficient to improve transfer.**
4. **Lexical/code novelty gains under failure or random replay are not indicative of algorithmic invention, as deterministic optimality gap metrics do not support a transfer advantage.**
5. **No evidence supports contention that replay (especially failure or random) improves transfer; on the contrary, it degrades transfer metrics compared to no replay.**

Summary Table: Rank by Transfer Performance

Rank	Condition	Mean Transfer Gap	Mean CVRPLIB Gap	Mean Synthetic Gap	Code Novelty (mean)	Replay Type
1	cvrp_no_replay	**0.1240**	**0.2188**	**0.0054**	0.7424	none
2	cvrp_failure_replay_compression	0.1347	0.2208	0.0270	0.7806	failure + compression
3	cvrp_random_replay	0.1366	0.2415	0.0054	0.7273	random
4	cvrp_failure_replay	0.1403	0.2483	0.0054	0.7627	failure

Overall, the no replay condition should be prioritized for CVRP transfer purposes. Replay strategies increase code novelty but do not yield better practical transfer performance according to held-out CVRPLIB and synthetic gaps.